

# Amazon Reviews 2023 – Appliances Dataset

## Exploratory Data Analysis (EDA) Report

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### **Project**

Enterprise Product Intelligence Platform

### **Report**

Data Understanding & Exploratory Data Analysis

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## 1. Executive Summary

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This report presents the Exploratory Data Analysis (EDA) conducted on the Amazon Reviews 2023 – Appliances dataset, obtained from the McAuley Lab Amazon Reviews 2023 collection. The objective of this analysis is to understand the structure, quality, completeness, and business characteristics of the dataset prior to implementing the Bronze-to-Silver ETL pipeline within the proposed Medallion Architecture.

The analysis covers both the Reviews and Metadata datasets, comprising 2,128,605 customer reviews and 94,327 product records. Processing was performed using Apache Spark, and multiple validation checks were carried out, including schema verification, relationship analysis, duplicate detection, missing value assessment, temporal trend analysis, product-level analysis, and customer review analysis.

Key findings indicate that the dataset possesses:

- Strong referential integrity between reviews and product metadata.
- High completeness across critical transactional attributes.
- Minimal duplicate records (1.06% of total reviews).
- Consistently high customer ratings, with an average review rating of 4.22 and an average product rating of 4.39.
- A very high verified purchase rate of 95.84%.
- Steady growth in customer engagement, peaking in 2021.

The findings presented in this report establish a reliable foundation for downstream data engineering tasks, including data cleaning, feature engineering, natural language processing, recommendation systems, business intelligence, and machine learning applications for the Appliances category.

## 2. Project Objectives

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The primary objective of this exploratory analysis is to develop a comprehensive understanding of the Amazon Appliances Reviews dataset before performing large-scale data transformation.

The analysis aims to:

- Understand the structure and characteristics of both the Reviews and Metadata datasets.
- Validate relationships between product metadata and customer reviews.
- Assess data quality, completeness, and consistency.
- Identify duplicate and missing records.
- Analyze customer engagement trends over time.
- Evaluate product popularity, store performance, and pricing behaviour.
- Study customer review characteristics, helpfulness, and sentiment.
- Define an appropriate stratified sampling strategy for downstream analytics.
- Establish a reliable foundation for the Bronze-to-Silver ETL pipeline.

The insights generated through this exploratory analysis will guide the preprocessing, integration, and feature engineering stages of the overall data engineering workflow.

### 3. Dataset Overview

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The Amazon Reviews 2023 Appliances dataset consists of two complementary datasets:

- Reviews Dataset – Contains customer review information, ratings, purchase verification, and review metadata.
- Metadata Dataset – Contains product information, pricing, categories, ratings, and descriptive attributes.

The Reviews dataset captures transactional customer interactions, while the Metadata dataset provides descriptive information for each product. Together, these datasets enable comprehensive customer behaviour analysis, product analytics, sentiment analysis, recommendation systems, and business intelligence applications.

#### 3.1 Reviews Dataset

| Metric        | Value     |
|---------------|-----------|
| Total Records | 2,128,605 |
| Total Columns | 10        |

#### 3.2 Metadata Dataset

| Metric        | Value  |
|---------------|--------|
| Total Records | 94,327 |
| Total Columns | 14     |

#### 3.3 Dataset Summary

| Dataset  | Records   | Columns |
|----------|-----------|---------|
| Reviews  | 2,128,605 | 10      |
| Metadata | 94,327    | 14      |

The combined datasets provide sufficient scale for enterprise-level analytical processing and distributed data engineering workflows on the Appliances product category.

### 4. Data Relationship Analysis

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Understanding the relationship between the Reviews and Metadata datasets is a critical prerequisite before integrating them into a unified analytical dataset. This analysis validates referential integrity, identifies appropriate business keys, and ensures that both datasets can be reliably merged during the ETL process.

## 4.1 Key Identification

The Metadata dataset uses `parent_asin` as the unique identifier for every product, making it the natural primary key.

The Reviews dataset does not contain a single-column primary key because multiple customers can review the same product and individual customers can review multiple products. To uniquely identify a review record, the following composite key was identified:

- `parent_asin`
- `user_id`
- `review_text`
- `review_timestamp`

This logical composite key was used throughout the exploratory analysis for duplicate detection and validation.

## 4.2 Relationship Validation

**Table 4.1 Relationship Validation**

| Validation Check                               | Result  |
|------------------------------------------------|---------|
| Duplicate <code>parent_asin</code> in Metadata | 0       |
| Missing <code>parent_asin</code> in Reviews    | 0       |
| Reviews without matching Metadata (Orphans)    | 0       |
| Metadata records without Reviews               | 8       |
| Merge Success Rate                             | 100.00% |

The analysis confirms that every customer review successfully references an existing product record. Only 8 metadata records were found without corresponding customer reviews, which is expected for newly listed or low-traffic products.

## 4.3 Key Observations

- Metadata maintains a unique `parent_asin` for every product.
- Reviews exhibit a one-to-many relationship with products.
- No orphan review records were detected.
- An inner join between Reviews and Metadata on `parent_asin` yields 2,128,605 merged records — a 100% merge success rate.

## 4.4 Summary

Relationship analysis confirms that the Reviews and Metadata datasets possess strong referential integrity and are structurally suitable for the Bronze-to-Silver transformation process.

## 5. Data Profiling

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Data profiling was conducted to understand the structure, completeness, uniqueness, and consistency of the merged dataset before performing any transformation or preprocessing.

### 5.1 Structural Overview

The merged dataset (Reviews joined with Metadata on parent\_asin) contains 2,128,605 rows. After feature engineering — timestamp conversion and the derivation of year, month, day, quarter, and week attributes — the dataset expands to 27 columns.

### 5.2 Cardinality Analysis

Table 5.1 Dataset Cardinality

| Attribute                     | Observation                |
|-------------------------------|----------------------------|
| Unique Products (parent_asin) | 94,319                     |
| Unique Users                  | 1,755,732                  |
| Unique Stores                 | 13,166                     |
| Main Categories               | 31                         |
| Total Reviews                 | 2,128,605                  |
| Relationship                  | One Product → Many Reviews |

### 5.3 Column-Level Profiling

Table 5.2 Column Profiling Summary

| Column        | Data Type | Missing % | Distinct Values |
|---------------|-----------|-----------|-----------------|
| parent_asin   | string    | 0.00      | 94,319          |
| asin          | string    | 0.00      | 104,237         |
| user_id       | string    | 0.00      | 1,755,732       |
| review_rating | double    | 0.00      | 5               |
| review_title  | string    | 0.00      | 999,079         |
| review_text   | string    | 0.00      | 1,840,887       |
| timestamp     | bigint    | 0.00      | 2,104,484       |

| Column            | Data Type | Missing % | Distinct Values |
|-------------------|-----------|-----------|-----------------|
| helpful_vote      | bigint    | 0.00      | 533             |
| verified_purchase | boolean   | 0.00      | 2               |
| price             | double    | 24.82     | 10,650          |
| store             | string    | 0.28      | 13,166          |
| main_category     | string    | 1.34      | 31              |

## 5.4 Summary

- All core transactional attributes (identifiers, ratings, timestamps, verification flags) are fully populated.
- Descriptive product attributes such as price, store, and main\_category show limited missingness.
- The dataset possesses well-defined schemas and stable business identifiers suitable for large-scale analytical processing.

## 6. Missing Value Analysis

Missing value analysis was performed on the merged dataset to evaluate completeness and identify attributes requiring preprocessing before downstream integration.

Table 6.1 Missing Values – Merged Dataset

| Column                                        | Missing Values | Missing % |
|-----------------------------------------------|----------------|-----------|
| price                                         | 528,346        | 24.82%    |
| main_category                                 | 28,494         | 1.34%     |
| store                                         | 6,064          | 0.28%     |
| parent_asin / asin / user_id                  | 0              | 0.00%     |
| review_rating / review_title / review_text    | 0              | 0.00%     |
| helpful_vote / verified_purchase              | 0              | 0.00%     |
| categories / features / description / details | 0              | 0.00%     |

### Observation

The merged dataset is highly complete across all transactional attributes. The most significant gap is in the price column (24.82% missing), which is notably higher than the equivalent field in other product categories, and is likely linked to a large proportion of third-party spare-part and accessory listings without list prices. main\_category and store show comparatively minor missingness.

### 6.1 Summary

The missing value assessment confirms that overall dataset quality is high. The identified missing values are largely confined to the price attribute and can be addressed during the Bronze-to-Silver ETL transformation — for example, through category-level price imputation or explicit flagging — without significantly affecting downstream analytics.

## 7. Duplicate Analysis

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Duplicate analysis was performed to identify redundant records that may impact aggregation, reporting, and machine learning model performance. Logical duplicates were identified using the composite key (user\_id, parent\_asin, review\_text, review\_timestamp).

**Table 7.1 Duplicate Analysis**

| Validation        | Count     |
|-------------------|-----------|
| Total Records     | 2,128,605 |
| Unique Records    | 2,105,948 |
| Duplicate Records | 22,657    |
| Duplicate %       | 1.06%     |

Considering the overall dataset size of 2,128,605 customer reviews, duplicate records account for close to 1% of the complete dataset. These records will be removed during the Bronze-to-Silver ETL pipeline to eliminate redundancy and improve analytical accuracy.

### 7.1 Summary

Duplicate analysis demonstrates that redundancy within the Appliances dataset is minimal and can be effectively resolved through standard deduplication techniques.

## 8. Data Quality Assessment

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A comprehensive data quality assessment was conducted to validate the consistency, reliability, and correctness of the dataset before downstream processing.

**Table 8.1 Data Quality Validation**

| Validation Check              | Result  |
|-------------------------------|---------|
| Invalid Ratings (outside 1–5) | 0       |
| Negative Prices               | 0       |
| Future-Dated Reviews          | 0       |
| Missing Prices                | 528,346 |

| Validation Check         | Result    |
|--------------------------|-----------|
| Timestamp Validation     | Passed    |
| Referential Integrity    | Passed    |
| Duplicate Detection      | Completed |
| Missing Value Assessment | Completed |

## 8.1 Key Findings

- No invalid customer ratings were detected — all values fall within the valid 1–5 range.
- No negative product prices were identified.
- No future-dated review timestamps were found; all timestamps are chronologically valid.
- Referential integrity between the Reviews and Metadata datasets is fully maintained.
- Duplicate records represent close to 1% of the overall dataset.

## 8.2 Summary

Overall, the Amazon Reviews 2023 Appliances dataset demonstrates strong data quality and is well suited for enterprise-scale ETL, business intelligence, natural language processing, and machine learning applications.

## 9. Year-wise Analysis

Year-wise analysis was performed to examine customer engagement, purchase behaviour, and seasonal trends over time. The analysis is based on review timestamps and provides insights into the historical evolution of customer activity within the Appliances category.

### 9.1 Review Distribution by Year



Figure 9.1: Annual customer review volume, 2000–2023.

Customer review activity increased steadily over time, with rapid growth occurring after 2013. Review volume reached its highest level in 2021 with 345,120 reviews, following strong growth through 2019 (261,004) and 2020 (311,251). A decline is visible in 2022 (299,487) and 2023 (125,053); the 2023 figure reflects partial-year data availability rather than reduced customer engagement.

## 9.2 Verified Purchase Trend

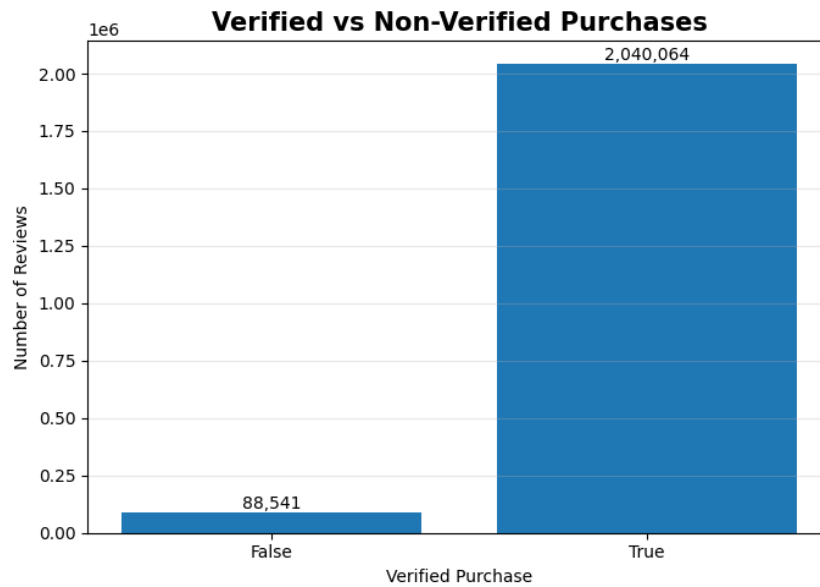


Figure 9.2: Verified vs. non-verified purchases across the dataset.

Verified purchases account for 2,040,064 reviews (95.84%), while only 88,541 reviews (4.16%) are non-verified. This exceptionally high verified purchase rate strengthens confidence in the reliability of customer sentiment and rating data for the Appliances category.

## 9.3 Monthly Review Trend

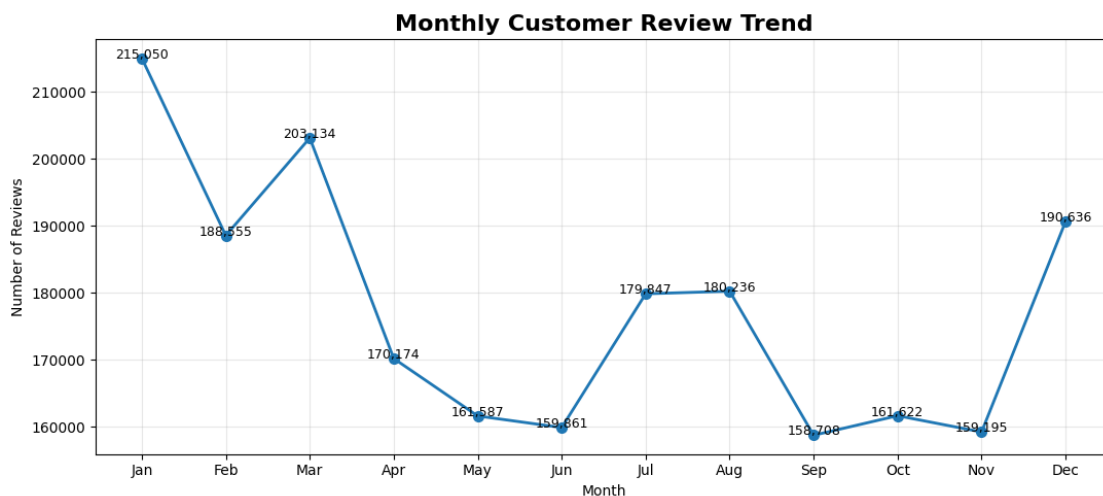


Figure 9.3: Monthly customer review trend, aggregated across all years.

Reviews are highest in January (215,050) and December (190,636), suggesting elevated activity around the holiday and new-year appliance purchase season. September records the lowest volume (158,708). Overall, monthly variation is moderate, indicating fairly consistent year-round engagement with some seasonal uplift.

## 9.4 Summary

- Customer engagement grew significantly between 2013 and 2021.
- Peak review activity occurred in 2021, with sustained high volume in 2019–2022.
- 95.84% of reviews originate from verified purchases.
- Monthly trends show a mild seasonal peak around January and December.

Overall, the temporal analysis indicates sustained marketplace growth, increasing customer participation, and a highly credible review base for the Appliances category.

## 10. Product Analysis

Product-level analysis was conducted to evaluate customer engagement, product popularity, pricing behaviour, and store performance within the Appliances marketplace.

### 10.1 Top Reviewed Products

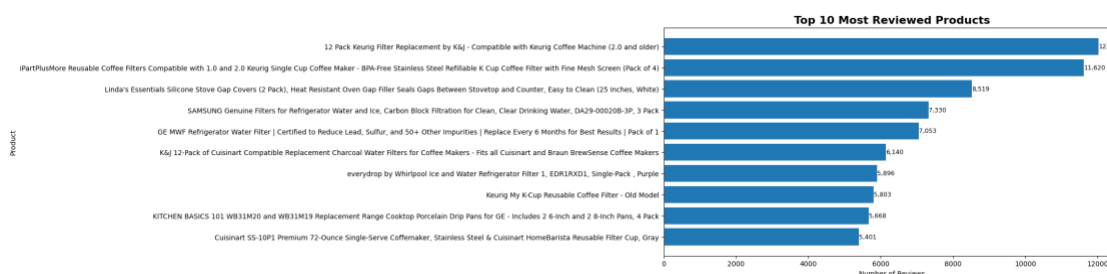


Figure 10.1: Top 10 products ranked by total customer reviews.

Customer reviews are heavily concentrated among a small number of highly popular products. The most reviewed product — a 12-Pack Keurig-compatible replacement filter — received 12,027 reviews, followed by a reusable coffee filter product (11,620) and silicone stove gap covers (8,519).

Table 10.1 Top 10 Most Reviewed Products

| Product                                      | Reviews |
|----------------------------------------------|---------|
| 12 Pack Keurig Filter Replacement (K&J)      | 12,027  |
| iPartPlusMore Reusable Coffee Filters        | 11,620  |
| Linda's Essentials Silicone Stove Gap Covers | 8,519   |
| SAMSUNG Genuine Refrigerator Water Filters   | 7,330   |
| GE MWF Refrigerator Water Filter             | 7,053   |
| K&J 12-Pack Cuisinart-Compatible Filters     | 6,140   |

| Product                                        | Reviews |
|------------------------------------------------|---------|
| everydrop by Whirlpool Ice & Water Filter      | 5,896   |
| Keurig My K-Cup Reusable Coffee Filter         | 5,803   |
| KITCHEN BASICS 101 WB31M20/WB31M19 Replacement | 5,668   |
| Cuisinart SS-10P1 Premium Single-Serve Filter  | 5,401   |

This behaviour follows the long-tail distribution commonly observed within e-commerce platforms. Notably, the Appliances category is dominated by consumable, replacement-part products (filters, gaskets, covers) rather than large durable appliances — a distinct pattern compared to categories like Sports & Outdoors.

## 10.2 Distribution of Average Product Ratings

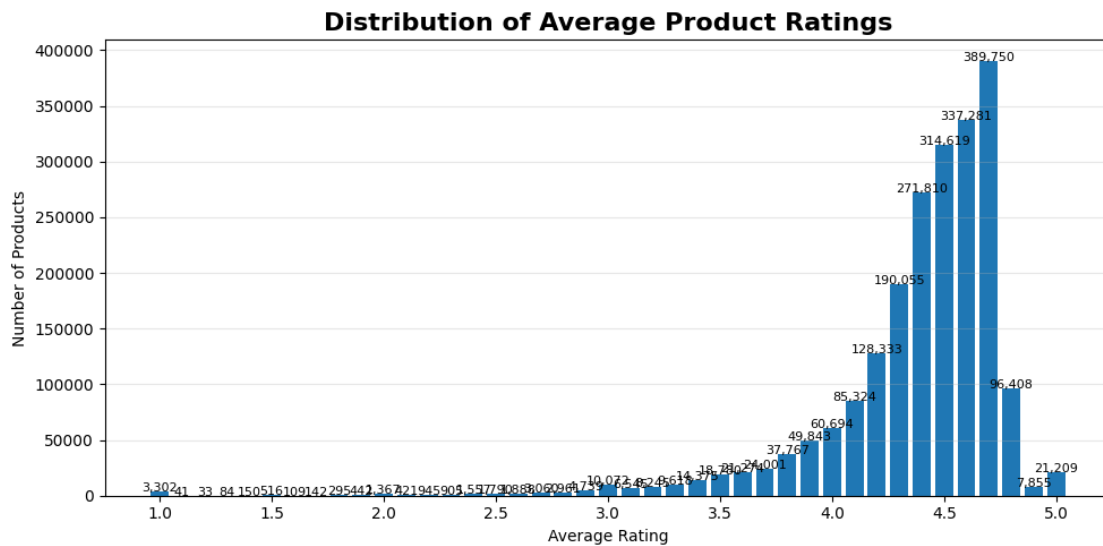


Figure 10.2: Distribution of average product ratings.

The overwhelming majority of products carry average ratings between 4 and 5 stars, consistent with an average product rating of 4.39 across the dataset. Very few products fall below 3 stars, resulting in a strongly left-skewed distribution indicative of high overall customer satisfaction.

## 10.3 Rating Count Distribution

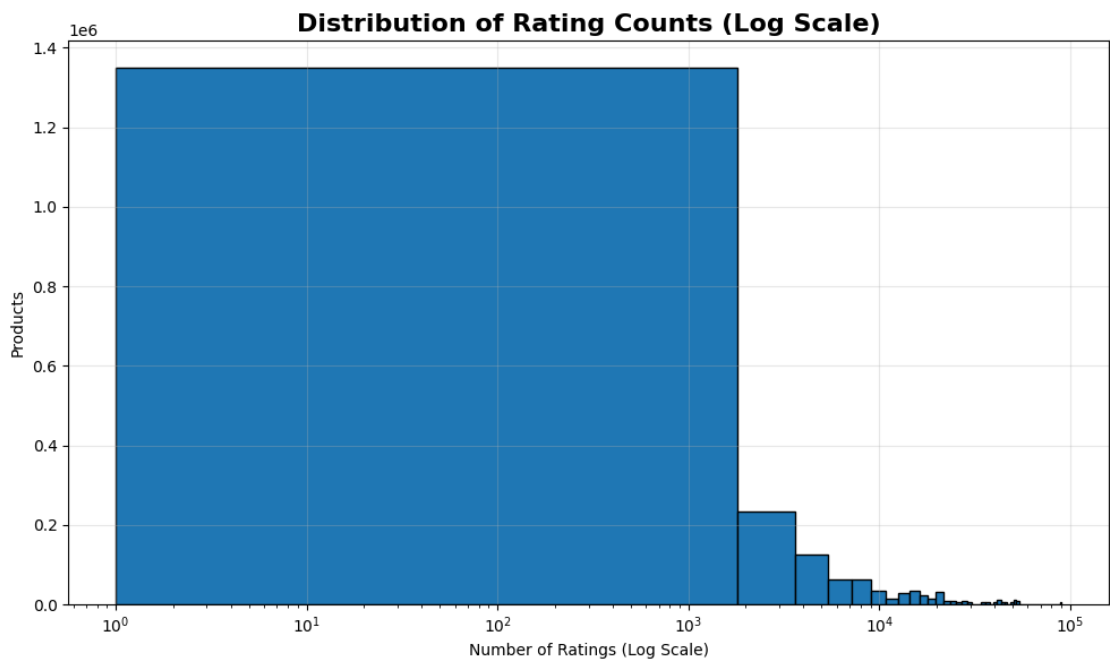


Figure 10.3: Distribution of the number of ratings per product (log scale).

Rating counts follow a highly skewed, long-tail pattern: most products accumulate a modest number of ratings, while a small subset of popular products accumulate disproportionately large rating counts.

10.4 Store Distribution

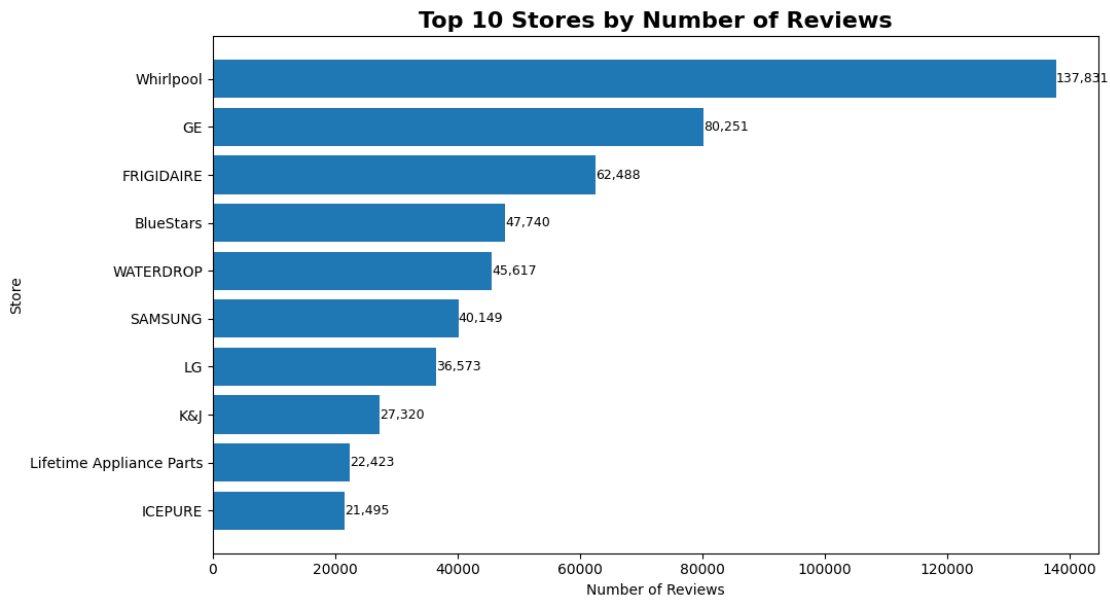


Figure 10.4: Top 10 stores ranked by number of reviews.

Table 10.2 Top 10 Stores by Review Count

| Store     | Reviews |
|-----------|---------|
| Whirlpool | 137,831 |

| Store                    | Reviews |
|--------------------------|---------|
| GE                       | 80,251  |
| FRIGIDAIRE               | 62,488  |
| BlueStars                | 47,740  |
| WATERDROP                | 45,617  |
| SAMSUNG                  | 40,149  |
| LG                       | 36,573  |
| K&J                      | 27,320  |
| Lifetime Appliance Parts | 22,423  |
| ICEPURE                  | 21,495  |

Major appliance manufacturers — Whirlpool, GE, Frigidaire, Samsung, and LG — dominate review volume alongside specialized replacement-parts brands such as BlueStars, Waterdrop, and ICEpure, reflecting the aftermarket-parts character of this category.

## 10.5 Product Price Distribution

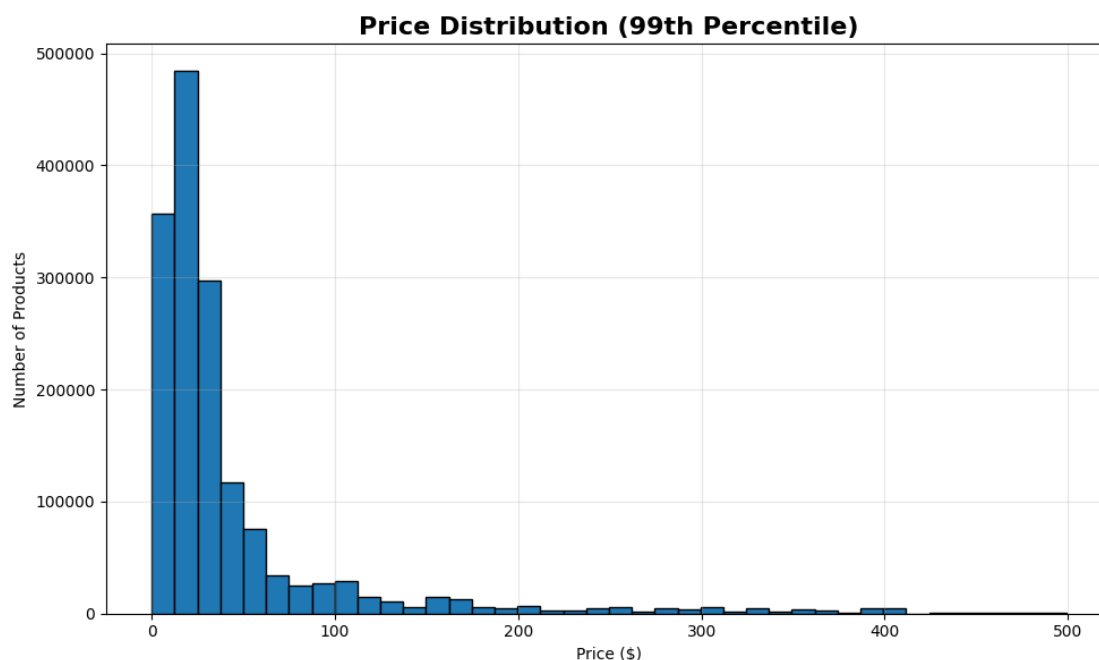


Figure 10.5: Distribution of product prices (99th percentile).

Product prices are strongly right-skewed, with an overall average price of \$54.87 and the large majority of listed products priced well below this figure. This is consistent with a marketplace dominated by low-cost consumable parts and accessories rather than full-size appliances.

## 10.6 Price vs. Average Rating

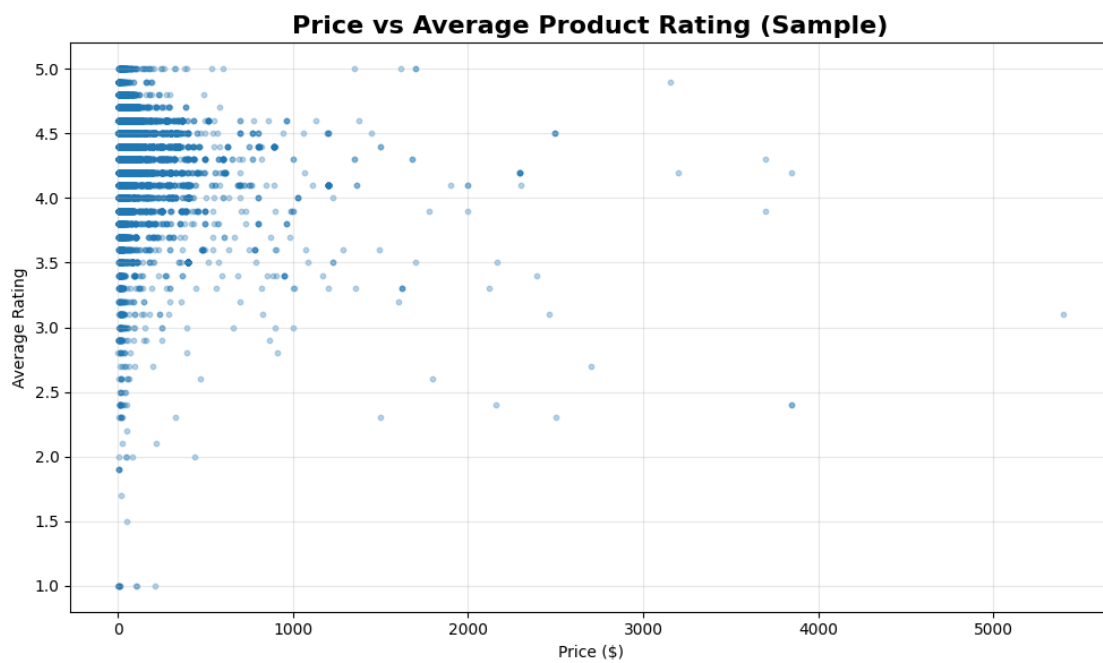


Figure 10.6: Price vs. average product rating (sampled).

No strong linear relationship is visually apparent between price and average product rating, suggesting that price is not a strong predictor of customer satisfaction in this category — a finding later confirmed by correlation analysis in Section 11.

## 10.7 Summary

- Customer engagement follows a long-tail distribution dominated by consumable replacement parts.
- Most products maintain average ratings above 4 stars.
- Major appliance brands and specialized parts suppliers dominate store-level engagement.
- Product pricing is concentrated at the lower end, with an overall average of \$54.87.
- Price shows little to no visible relationship with product rating.

## 11. Review Analysis

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Review-level analysis was conducted to understand customer feedback characteristics, review usefulness, rating behaviour, and attribute relationships. These insights provide valuable information for downstream Natural Language Processing (NLP), sentiment analysis, and recommendation system development.

### 11.1 Review Rating Distribution

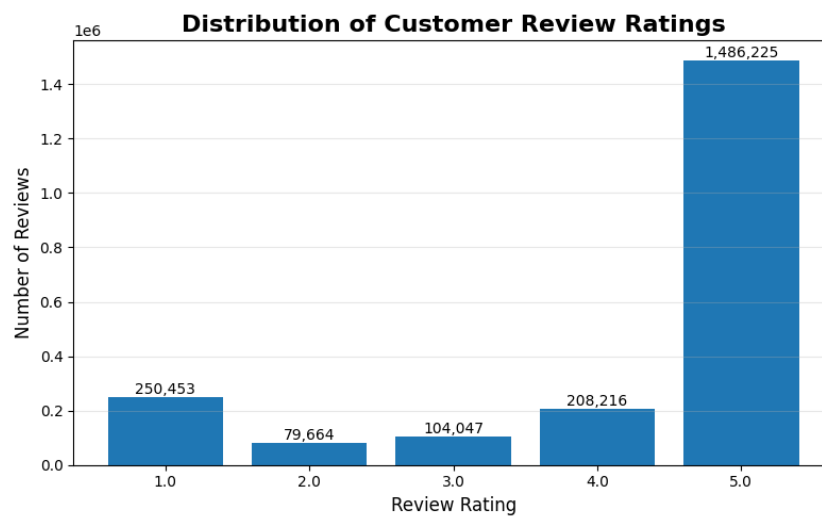


Figure 11.1: Distribution of customer review ratings.

Review ratings are heavily skewed toward the positive end of the scale, consistent with the average review rating of 4.22 across the full dataset of 2,128,605 reviews.

## 11.2 Review Length Distribution

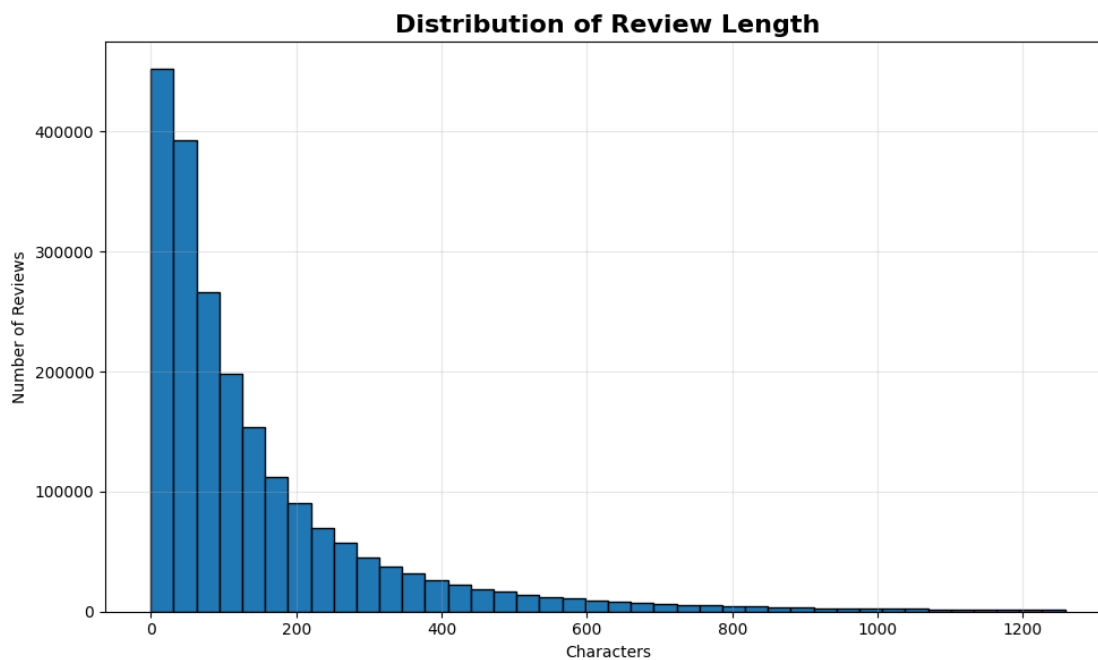


Figure 11.2: Distribution of customer review lengths (99th percentile).

The majority of reviews are relatively short, with a right-skewed distribution indicating that concise reviews dominate the dataset, while a smaller proportion of reviews contain detailed customer experiences.

## 11.3 Helpful Vote Distribution

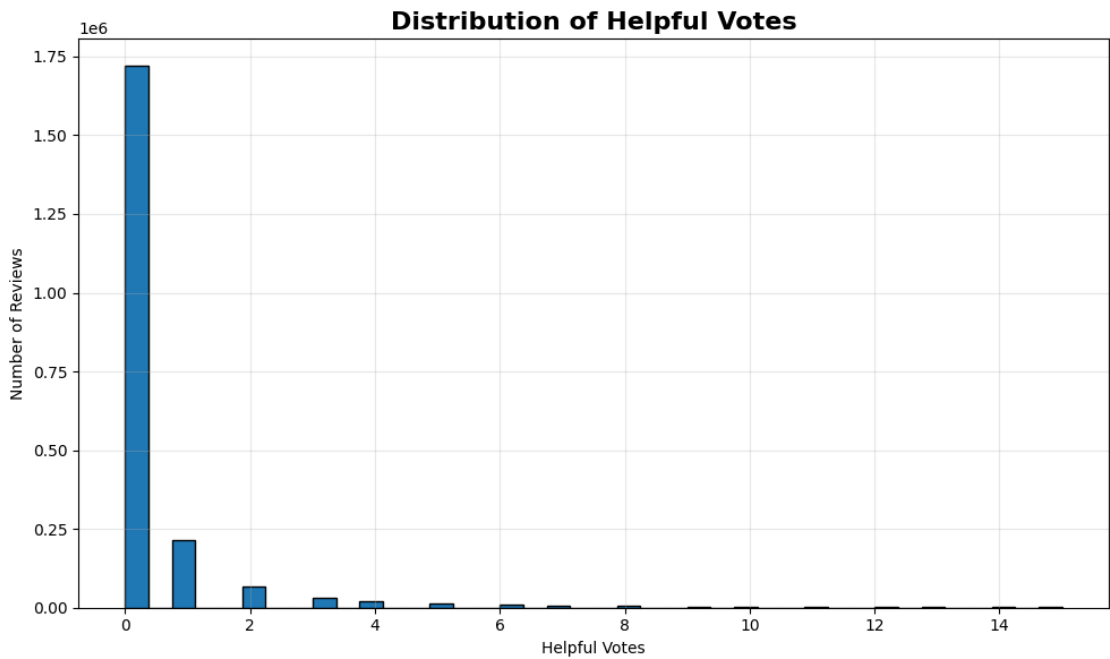


Figure 11.3: Distribution of helpful votes received by customer reviews (99th percentile).

Most reviews receive few or no helpful votes, while only a limited number accumulate a large number of helpful votes. This long-tail behaviour is commonly observed in user-generated content platforms and suggests that highly-voted reviews could be prioritized for recommendation and summarization use cases.

11.4 Correlation Analysis

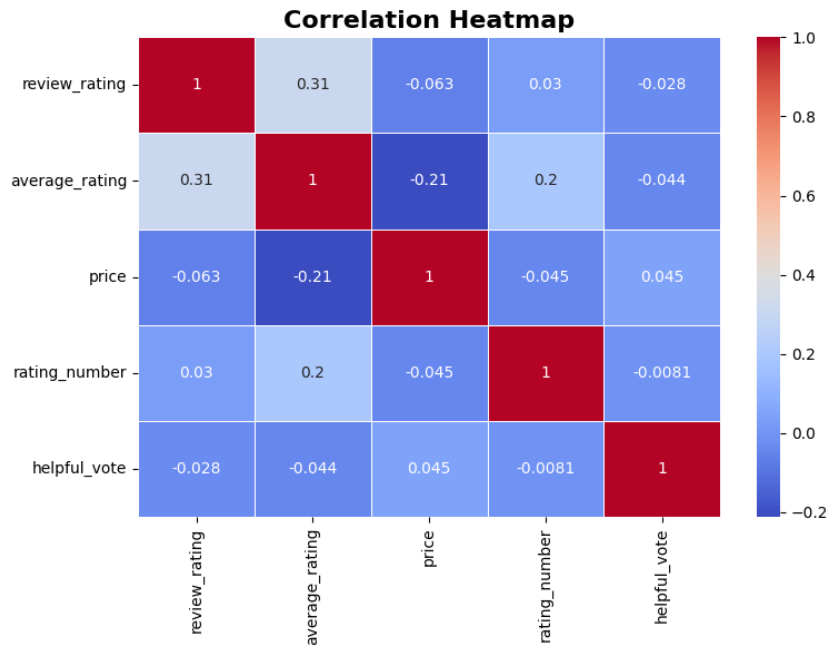


Figure 11.4: Correlation heatmap of review\_rating, average\_rating, price, rating\_number, and helpful\_vote.

Correlation analysis shows only weak relationships among numeric attributes — most notably, price exhibits a weak correlation with both review\_rating and average\_rating, confirming that pricing has limited influence on perceived product quality in this category.

11.5 Summary

- Customer reviews are generally concise but information-rich.
- Helpful vote distribution follows a long-tail pattern.
- Positive ratings dominate the dataset.
- Numeric attributes show weak inter-correlation, indicating rating behaviour is driven by factors beyond price alone.
- The review corpus is well suited for Natural Language Processing and sentiment analysis applications.

12. Sampling Strategy

Processing the complete dataset of 2,128,605 reviews for iterative exploratory analysis and visualization is computationally expensive and unnecessary during the initial development phase. A stratified sampling strategy was therefore designed to support exploratory analysis while preserving the temporal and distributional characteristics of the original dataset.

12.1 Population Validation

An initial check of minimum reviews per year across the full 2000–2023 range showed that early years contained too few records (as low as 1 review) to support reliable stratified sampling. The sampling population was therefore restricted to the 2014–2023 window, where every year contains at least 77,353 reviews — comfortably exceeding the 10,000-review minimum threshold required for the sampling design.

12.2 Stratified Sampling Design

- Sampling Method: Stratified random sampling, partitioned by review year.
- Sampling Window: 2014–2023 (10 years).
- Rows per Year: 10,000 (selected via a random row-number window function, seed = 42).
- Final Sample Size: 100,000 records.

Table 12.1 Sampling Summary

| Metric                          | Value                      |
|---------------------------------|----------------------------|
| Population Dataset              | 2,128,605                  |
| Sampling Population (2014–2023) | 1,845,384                  |
| Final Sample Size               | 100,000                    |
| Sampling Technique              | Stratified Random Sampling |

| Metric            | Value       |
|-------------------|-------------|
| Sampling Variable | Review Year |
| Rows per Year     | 10,000      |
| Random Seed       | 42          |

The resulting 100,000-record stratified sample provides equal yearly representation across the 2014–2023 window and was used to support faster, iterative visual EDA (Section 13) while the complete dataset will be processed during the production ETL pipeline.

## 13. Sample-Based Exploratory Analysis

Using the 100,000-record stratified sample, a deeper visual exploration was conducted, including category composition, sentiment classification, and text-based analysis.

### 13.1 Top Main Categories

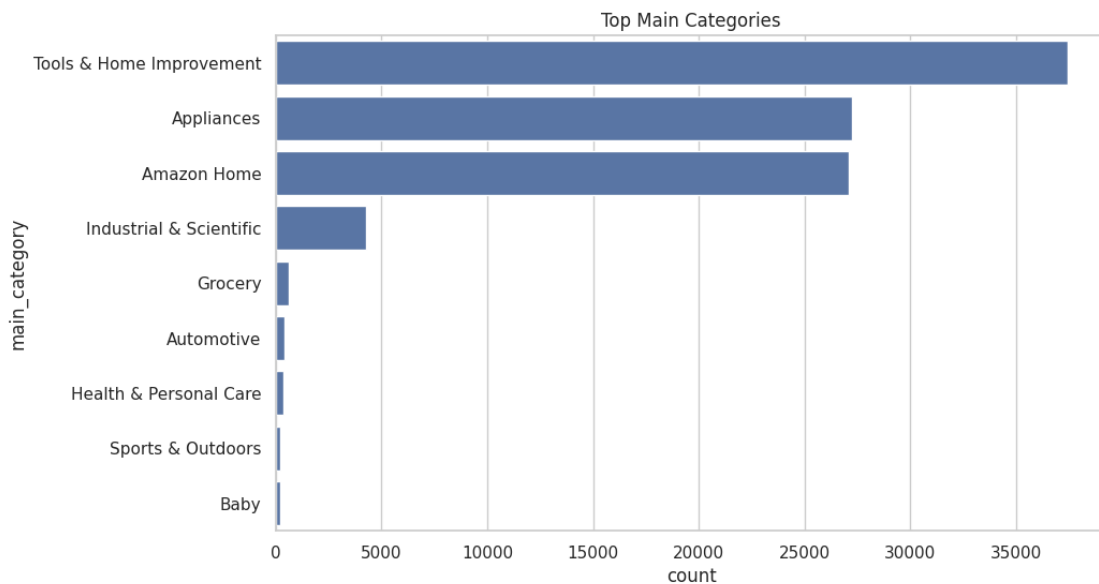


Figure 13.1: Top main categories represented in the sampled dataset.

While the dataset is sourced from the Appliances category, products also carry cross-listed `main_category` labels (such as Tools & Home Improvement), reflecting Amazon's overlapping category taxonomy for parts and accessories.

### 13.2 Sentiment Classification

Sentiment labels were derived from `review_rating`: ratings of 4–5 were classified as Positive, a rating of 3 as Neutral, and ratings of 1–2 as Negative.

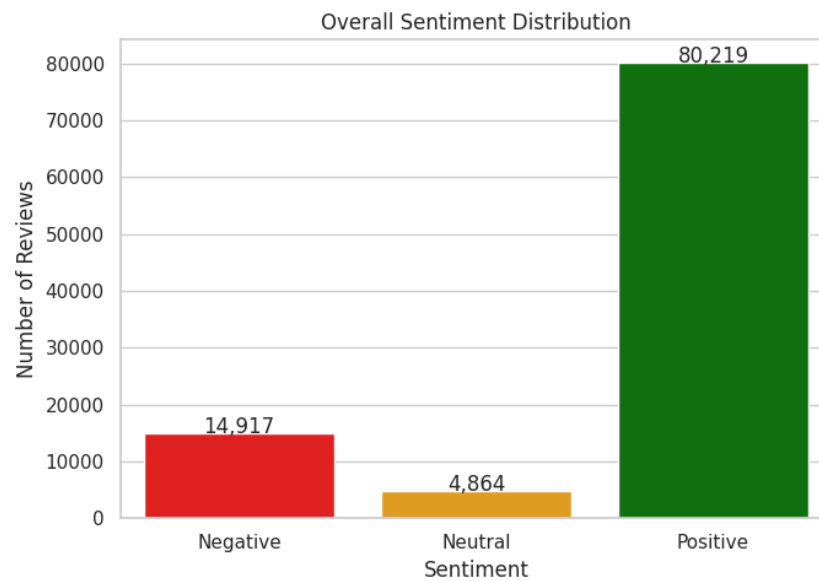


Figure 13.2: Overall sentiment distribution (sampled dataset).

Positive sentiment dominates the sample, consistent with the high average review rating observed across the full population.

### 13.3 Sentiment Trend by Year



Figure 13.3: Sentiment distribution by year (sampled dataset, 2014–2023).

Positive sentiment consistently leads across every sampled year, with negative and neutral sentiment remaining a small, stable minority — indicating sustained customer satisfaction over time rather than a one-off trend.

### 13.4 Frequent Review Terms

Frequent terms center on product functionality, fit, and satisfaction — consistent with a marketplace centered on functional replacement parts and accessories rather than aspirational or lifestyle products.

- The stratified sample preserves the rating, sentiment, and temporal characteristics of the full population.
- Positive sentiment dominates consistently across all sampled years.
- Frequent review terms confirm a functional, parts-and-accessories-driven customer base.

The exploratory analysis generated several important business insights:

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- Product price shows only a weak relationship with customer ratings, indicating that product quality and fit — not price — are the primary satisfaction drivers.
- Review helpfulness is concentrated among a small number of detailed, highly-voted reviews.
- The dataset demonstrates strong data quality and excellent suitability for large-scale analytics.

## 15. Recommendations

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Based on the exploratory analysis, the following recommendations are proposed:

- Remove duplicate review records (1.06% of the dataset) during the Bronze-to-Silver ETL transformation.
- Address missing price values (24.82%) using category-level or store-level imputation, or explicit "price unavailable" flagging for downstream models.
- Standardize store names and category labels to improve reporting consistency, given the cross-listing of Appliances products under other main\_category values.
- Prioritize verified purchase reviews during sentiment modelling and recommendation system development.
- Generate derived features such as review length, review age, and product popularity rank to enhance downstream analytical models.
- Build a separate analytical view for replacement-parts / consumables versus full appliances, given their materially different price and review patterns.
- Continue using the stratified 2014–2023 sample for rapid iteration, reserving full-dataset processing for production Spark pipelines.
- Utilize distributed processing frameworks such as Apache Spark for large-scale feature engineering and machine learning workflows.

## 16. Conclusion

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The exploratory analysis confirms that the Amazon Reviews 2023 Appliances dataset is a high-quality, enterprise-scale dataset suitable for advanced analytical applications.

The dataset demonstrates excellent structural consistency, strong referential integrity, minimal duplication, and high completeness across critical transactional attributes. Customer reviews indicate consistently positive sentiment, while a 95.84% verified purchase rate and stable, high average ratings further strengthen confidence in the reliability of the data. The category's distinctive long-tail concentration around consumable replacement parts — rather than full appliances — represents an important structural characteristic to carry forward into feature engineering and business intelligence design.

The findings presented in this report establish a solid foundation for subsequent stages of the project, including data preprocessing, feature engineering, Medallion Architecture implementation, Natural Language Processing, recommendation systems, and business intelligence dashboards.

## 17. Appendix

### Appendix A: Dataset Statistics

| Dataset                           | Records   | Columns |
|-----------------------------------|-----------|---------|
| Reviews                           | 2,128,605 | 10      |
| Metadata                          | 94,327    | 14      |
| Merged (post feature engineering) | 2,128,605 | 27      |

### Appendix B: Figures

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- Figure 13.4 – Frequent Review Terms Word Cloud (Sample)

### Appendix C: Abbreviations

| Abbreviation | Description                 |
|--------------|-----------------------------|
| EDA          | Exploratory Data Analysis   |
| ETL          | Extract, Transform, Load    |
| NLP          | Natural Language Processing |

| Abbreviation | Description                                  |
|--------------|----------------------------------------------|
| BI           | Business Intelligence                        |
| ASIN         | Amazon Standard Identification Number        |
| PGCP         | Post Graduate Certificate Program            |
| CDAC         | Centre for Development of Advanced Computing |